Simple Electricity Bill Calculator

1. Problem Statement

Calculate a household electricity bill based on units consumed. Charge Rs. 2 per unit for the first 100 units and Rs. 3 per unit for units above 100.

2. Objectives

1. Read consumer name and units consumed.
2. Calculate bill using two consumption slabs.
3. Use conditional statements.
4. Display the electricity bill.

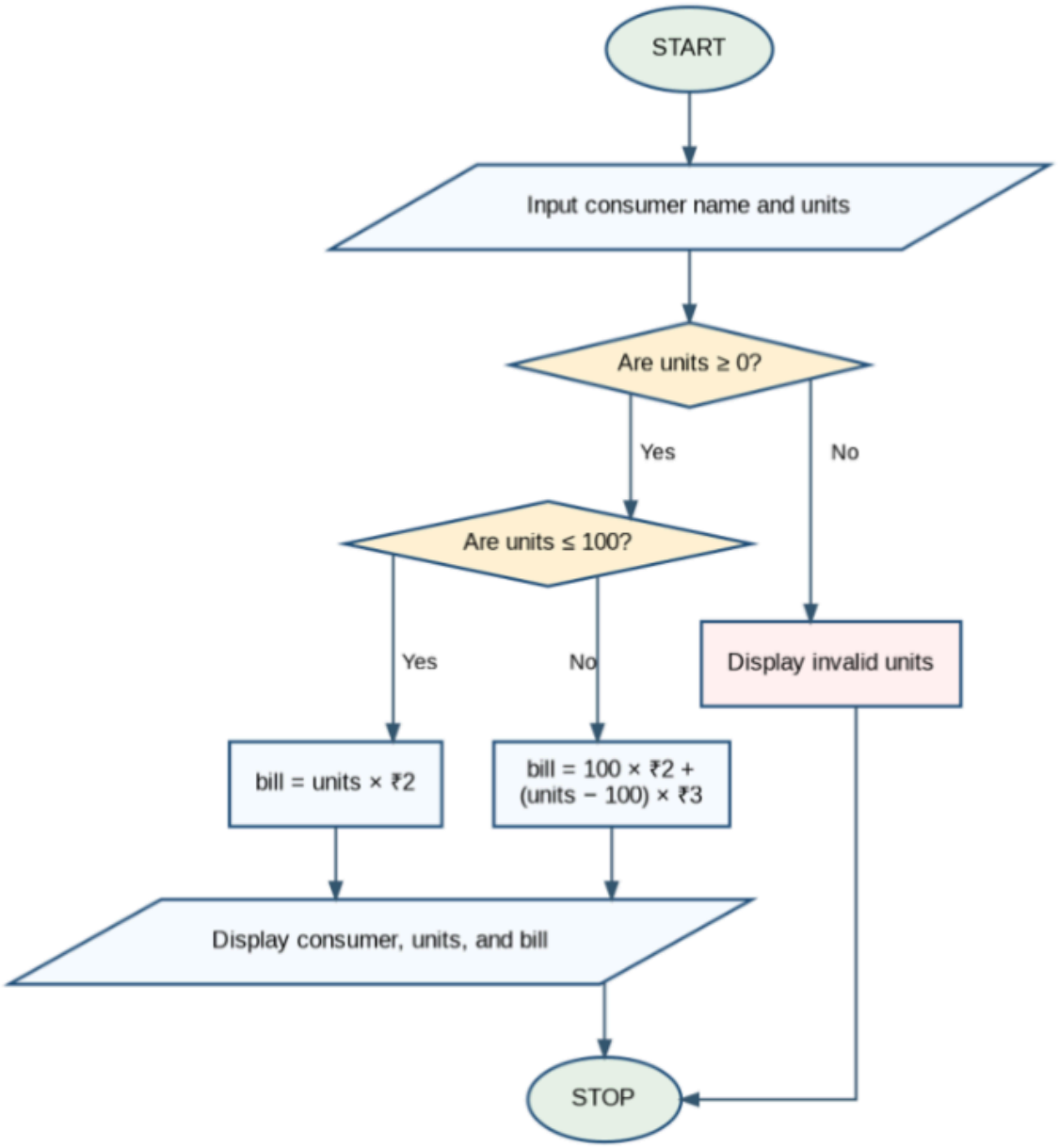
3. Assumptions

1. Units consumed cannot be negative.
2. First 100 units cost Rs. 2 per unit.
3. Units above 100 cost Rs. 3 per unit.
4. No fixed charge or tax is included.

4. Algorithm

1. Start.
2. Read consumer name and units consumed.
3. If units are up to 100, calculate bill as units × 2.
4. Otherwise, calculate bill as 100 × 2 plus remaining units × 3.
5. Display consumer name, units, and bill amount.
6. Stop.

5. Flowchart



None

6. Pseudocode

None

```
INPUT name, units
```

```
IF units <= 100 THEN
    bill = units * 2
ELSE
    bill = 100 * 2 + (units - 100) * 3

DISPLAY name, units, bill
```

8. Test Case and Expected Output

Input:

```
None
Consumer = Ravi
Units = 150
```

Calculation:

Bill = (100 × 2) + (50 × 3)

Bill = 200 + 150 = Rs. 350

Expected Output:

```
None
----- ELECTRICITY BILL -----
Consumer: Ravi
Units consumed: 150.0
Bill amount: Rs. 350.00
```

9. Conclusion

The program calculates an electricity bill using slab rates and a conditional statement.